

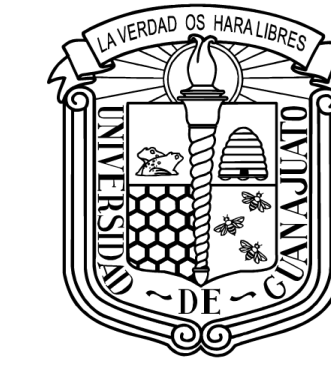


# Genetic Algorithm Search for Stable Vacua in Type IIB String Theory: A Computational Approach

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## ABSTRACT

The exploration of high-dimensional, non-linear potential landscapes is a central computational challenge. In string theory the search for stable vacua can be treated as an optimisation problem in order to include metaheuristic algorithms. In this work, a simple Genetic Algorithm (GA) is tailored to the optimisation of a type IIB string theory potential with torsion, extending the analysis of a previous work.

## OBJECTIVE

Extending previous work done in [1] using GA [2] instead of Simulate Annealing. This approach aims to use a population based algorithm instead of a single-point based like the original for further exploration and thus, further finding of stable configurations.

## INTRODUCTION

The application of computational intelligence to string theory has grown steadily over the past decade, with metaheuristics playing an increasingly central role in navigating the enormous space of possible vacuum configurations.

In previous works [3,4] GAs demonstrated being useful for vacua search in the field of string theory, searching configurations in less time and with less resources than using brute force and discovering new stable vacua configurations.

## METHODOLOGY

This study implements 4 penalisation elements in a fitness function which is intended to be minimised.

The penalisation elements are *descriptors* from the potential effective function

$$V_{eff} = \frac{A_{H_3} s}{\tau^3} + \frac{A_{F_3}}{s \tau^3} + \frac{A_{F_5}}{\tau^4} + \frac{A_3 \% N_3}{\tau^3} + \frac{A_{\hat{D}5}}{s^{1/2} \tau^{5/2}}$$

The descriptors are

$$f_{grad} = |\nabla \% V_{eff}|^2$$

$$p_{pos} = |\% V_{eff}| - \% V_{eff}$$

$$p_{tr} = |tr(\mathbf{H})| - tr(\mathbf{H})$$

$$p_{disc} = |\Delta| - \Delta \text{ with } \Delta = \det(\mathbf{H}) - \frac{1}{4}(tr \mathbf{H})^2$$

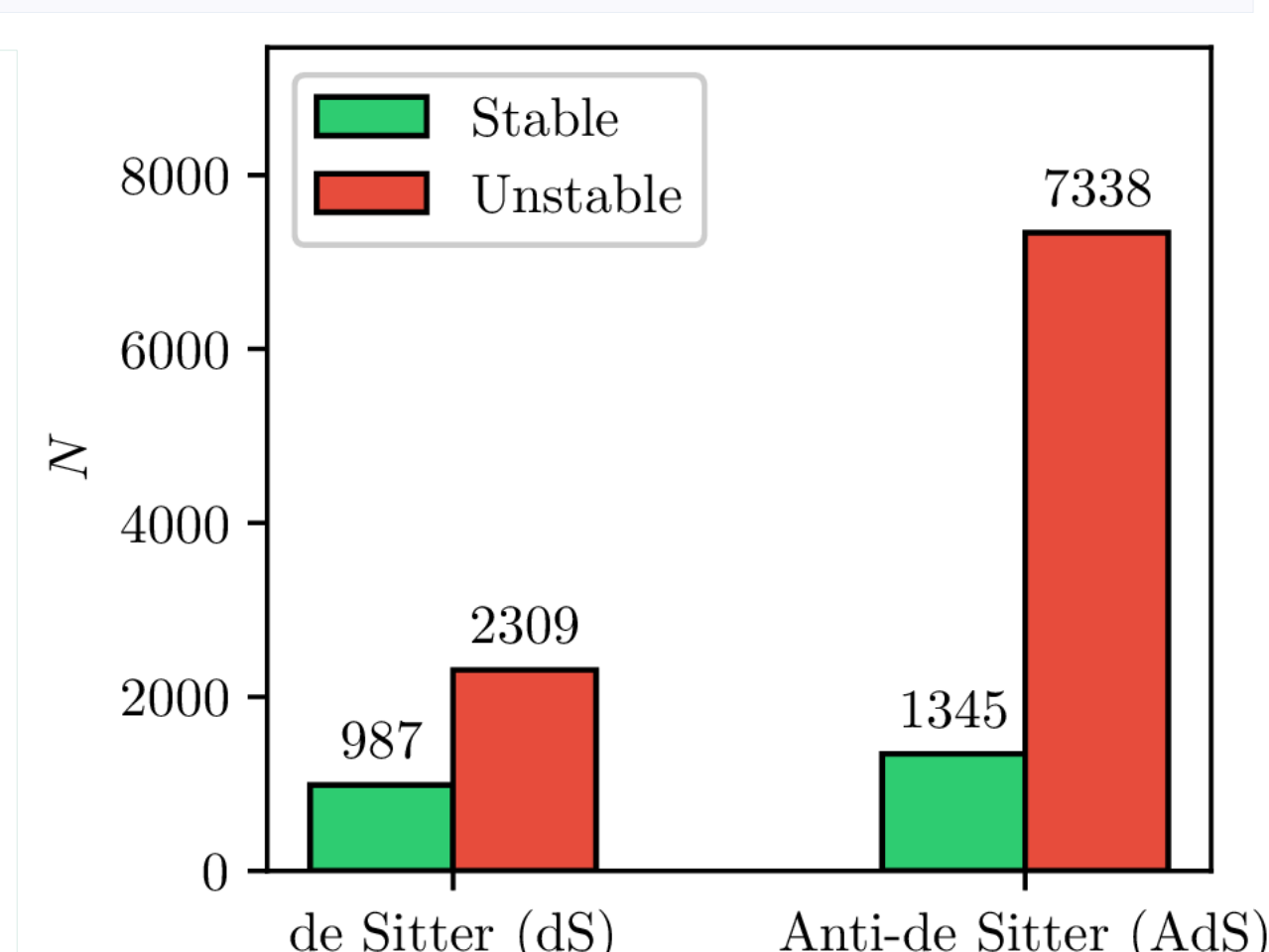
For a basic GA from MEALPY modulus and collect the results.

Finally, a **grid** search will be implemented to find a better model which is able to find higher amount of stable de Sitter vacua and the smallest fitness and potential effective values.

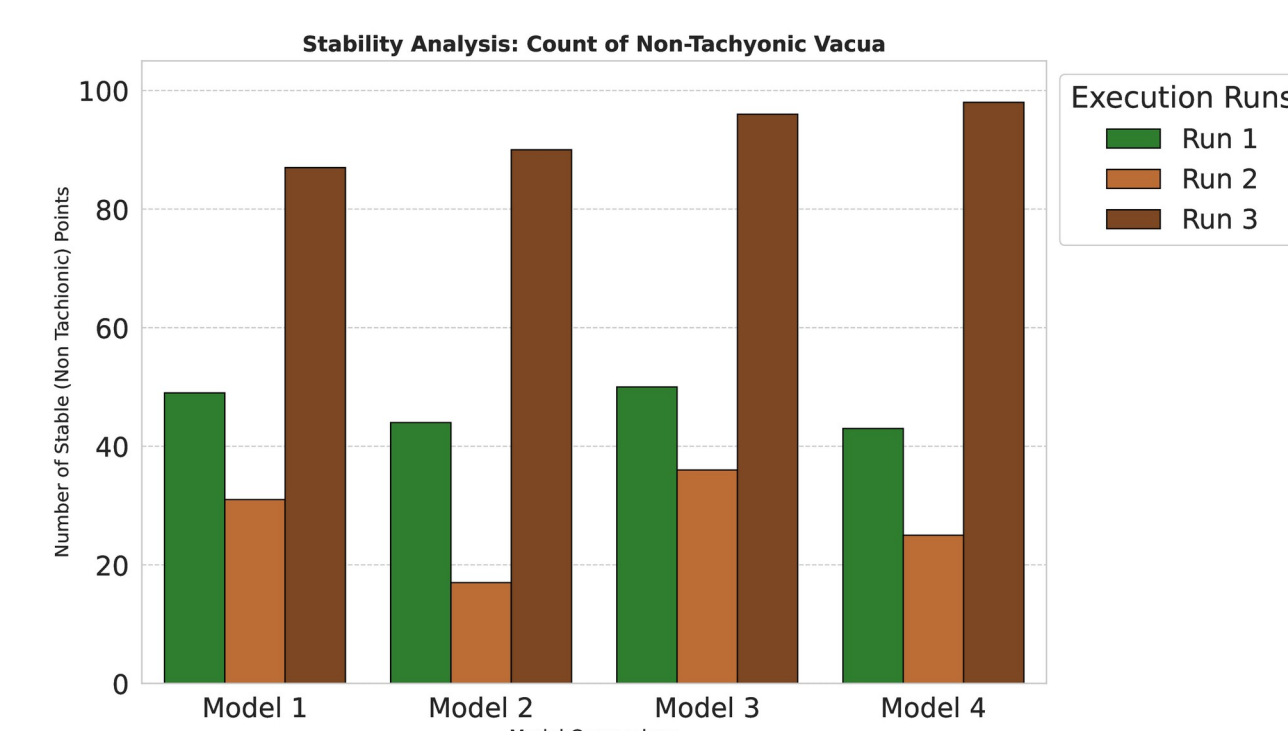
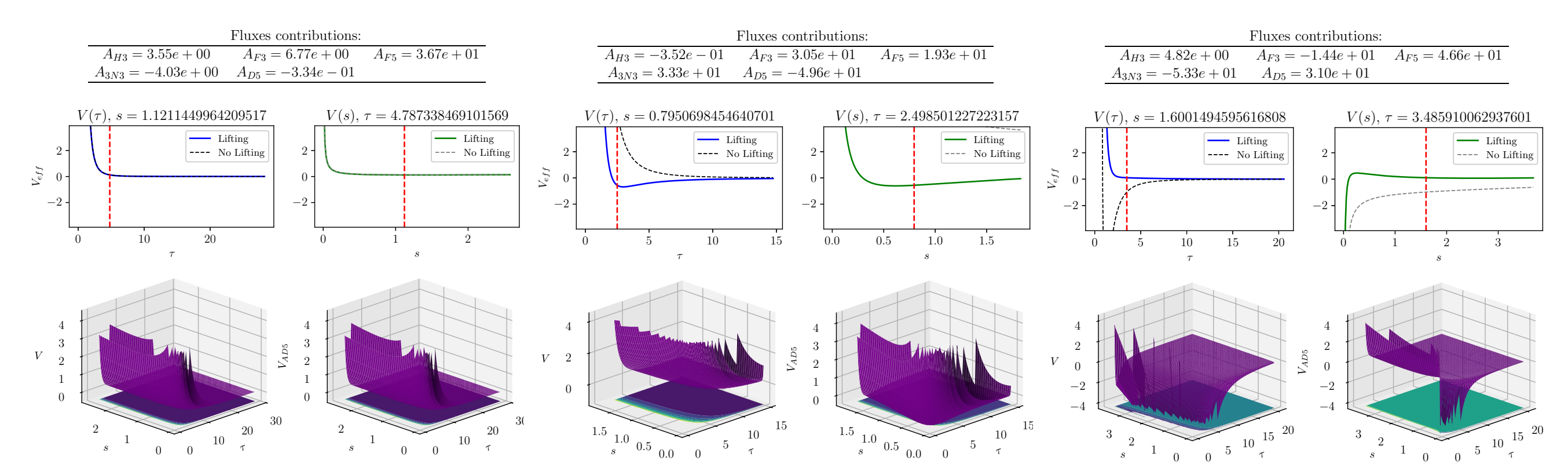
## RESULTS

After the search, **11 979** vacua were found: **987** dS stables and **1 345** AdS stables.

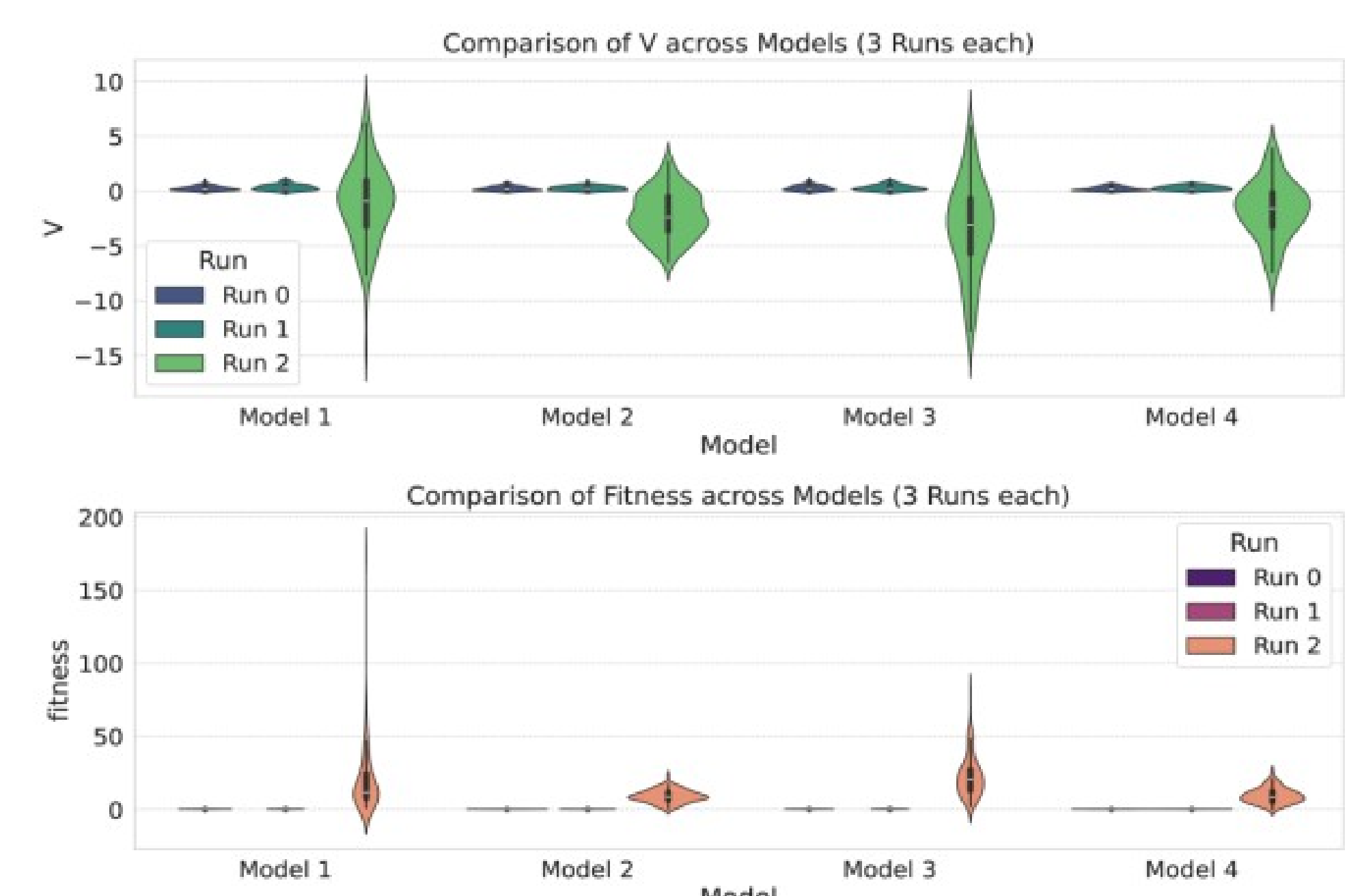
The term  $A_3 N_3$  was found to be consistently definite negative even though the bounds were  $|A_3 N_3| < 100$ .



Some vacua were found to have interesting behaviours after lifting such as No visible change, Massive lowering, Only s affected.



After the grid, it was found that, from 4 models, the best one was the one with 50 generations, 20 individuals, 0.5 of cross probability and 0.6 of mutation probability.



## CONCLUSIONS

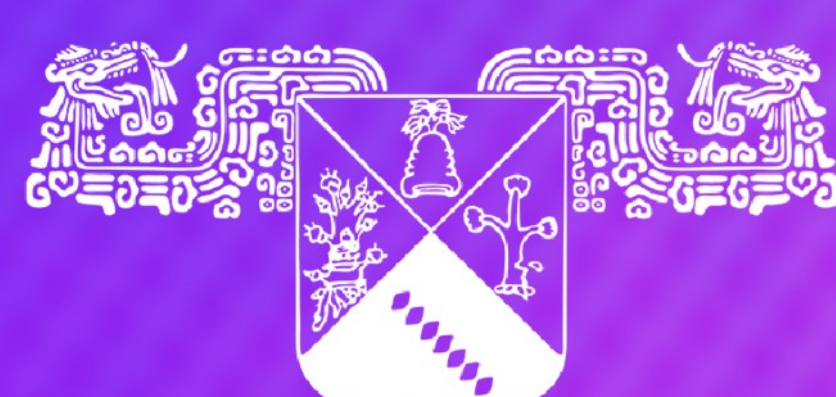
In this work 11 979 critical points were discovered, confirming stable de Sitter vacua. Systematic comparison of non-lifted vs. lifted potentials revealed interesting behaviours. The coefficient  $A_3 N_3$  naturally converged to negative values, matching physical expectations. Some numerical instabilities remain though. Grid analysis shows GA tuning improves results. Future work will test other nature-inspired algorithms.

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